

24/01/25

* Formulas for Sets

- (i) $n(A \cup B) = n(A) + n(B) - n(A \cap B)$ (overlapping)
- (ii) $n(A \cup B) = n(A) + n(B)$ (disjoint sets)
- (iii) $n(A) = n(A \cup B) + n(A \cap B) - n(B)$
- (iv) $n(B) = n(A \cup B) + n(A \cap B) - n(A)$
- (v) $n(A \cap B) = n(A) + n(B) - n(A \cup B)$
- (vi) $n(A \cup B) = n(A - B) + n(B - A) + n(A \cap B)$
- (vii) $n(U) = n(A) + n(B) - n(A \cap B) + n((A \cup B)^c)$
- (viii) $n(A - B) = n(A \cap B) - n(B)$
- (ix) $n(A - B) = n(A) - n(A \cap B)$
- (x) $n(A^c) = n(U) - n(A)$
- (xi) $n(A \cup B \cup C) = n(A) + n(B) + n(C) - n(A \cap B) - n(B \cap C) - n(C \cap A) + n(A \cap B \cap C)$

* Set theory Notation

Symbol Symbol Name

$\{ \}$ $\longrightarrow$ Set

$\cup$ $\longrightarrow$ Union

$\cap$ $\longrightarrow$ Intersection

$\subseteq$ $\longrightarrow$ Subset

$\subset$ $\longrightarrow$ Proper Subset

$\not\subset$ $\longrightarrow$ Not a subset

$\supseteq$ $\longrightarrow$ Superset

$\supset$ $\longrightarrow$ Proper Superset

Symbol	Symbol Name
$\not\supset$	Not a superset
A'	Complement
$-$	Difference
$\emptyset$	Empty Set
$P(A)$	Power Set
$ A $	Cardinality or Cardinal number

Q- Among a group of students, 50 played Cricket, 50 played Hockey and 40 played Valley ball. 15 played both Cricket and hockey, 20 played both hockey and Valley ball, 15 played both Cricket and Valley ball and 10 played all three. If every student played at least one game, Find number of student.

Sol = $n(C) = 50$, $n(H) = 50$, $n(V) = 40$
 $n(C \cap H) = 15$, $n(H \cap V) = 20$
 $n(C \cap V) = 15$, $n(C \cap H \cap V) = 10$

No of students who played at least one game

$$\begin{aligned}
 n(C \cup H \cup V) &= n(C) + n(H) + n(V) - n(C \cap H) - n(H \cap V) - n(C \cap V) + n(C \cap H \cap V) \\
 &= 50 + 50 + 40 - 15 - 20 - 15 + 10 \\
 &= 150 - 50 \\
 &= 100
 \end{aligned}$$

Q- In a class of 100 students, 35 like science and 45 like math, 10 like both. How many like either of them and how many like neither

Sol=

$$n(T) = 100$$

$$n(S) = 35$$

$$n(M) = 45$$

$$n(S \cap M) = 10$$

No of students who like either of them

$$n(S \cup M) = n(S) + n(M) - n(S \cap M)$$

$$= 35 + 45 - 10$$

$$= 70 - 10$$

$$[n(S \cup M) = 70]_{//}$$

No of students who like neither =

$$n(T) - n(S \cup M)$$

$$= 100 - 70$$

$$= 30_{//}$$

Q- There are 30 students in a class. Among them, 8 students are learning both English and French. A total of 18 students are learning English. If every student is learning at least one language, How many students are learning French in total.

Sol =

$$n(T) = 30$$

$$n(E) = 18$$

$$n(E \cap F) = 8$$

$$n(T) \text{ or } n(E \cup F) = n(E) + n(F) - n(E \cap F)$$

$$30 = 18 + n(F) - 8$$

$$n(F) = 30 - 18 + 8$$

$$[n(F) = 20]_{\parallel}$$

Total student learning French = 20_{||}

$$\begin{aligned} \text{Those learning only French} &= n(F) - n(E \cap F) \\ &= 20 - 8 \\ &= 12_{\parallel} \end{aligned}$$

$$\begin{aligned} \text{Those learning only English} &= n(E) - n(E \cap F) \\ &= 18 - 8 \\ &= 10_{\parallel} \end{aligned}$$